MILITARY SPECIFICATION

KIT, INSTALLATION OF COMMUNICATIONS EQUIPMENT IN 1-1/4 TON COMMERCIAL TRUCK

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification establishes the requirements for kit, installation of communication equipment in 1-1/4 ton commercial truck (hereafter referred to as kit assembly). The kit assembly shall be a lightweight, easily assembled and installed unit used to provide for a standardized installation of Army communications equipment in a 1-1/4 ton commercial truck. (see 6.1 and 6.1.1.)

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

QQ-S-781	Strapping, Steel, Flat and Seals
PPP-B-20	Bag, Cotton, Mailing
PPP-B-621	Boxes, Wood, Nailed and Lock-Corner
PPP-B-636	Boxes, Shipping, Fiberboard
PPP-C-843	Cushioning Material, Cellulosic

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PPP-F-320	Fiberboard, Corrugated and Solid Sheet Stock (Container Grade) and Cut Shapes
PPP-P-291	Paperboard, Wrapping, Cushioning
PPP-T-97	Tape, Pressure-Sensitive Adhesive Paper, Filament Reinforced

MILITARY

MIL-P-116	Preservation-Packaging, Methods of
MIL-W-6858	Welding, Resistance, Aluminum, Magnesium, Non-hardening Steels or Alloys, Nickel Alloys, Heat-resisting Alloys, and Titanium Alloys, Spot and Seam
MIL-M-13231	Marking of Electronic Items
MIL-F-14072	Finishes for Ground Signal Equipment
MIL-P-19644	Plastic Molding Material (Polystyrene Foam, Expanded Bead)
MIL-P-26514	Polyurethane Foam, Rigid or Plastic, for Packaging
MIL-W-45205	Welding, Gas Metal-arc and Gas Tungsten-arc, Aluminum Alloys, Readily Weldable For Structures, Excluding Armor

STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-252	Wired Equipment, Classification of Visual and Mechanical Defects
MIL-STD-454	Standard Requirements for Electronic Equipment
MIL-STD-810	Environmental Test Methods
MIL-STD-1472	Human Engineering Design Criteria for Military Systems, Equipment and Facilities

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DRAWINGS

ELECTRONICS COMMAND

DL-SC-B-48521	Antenna Bracket
DL-SC-B-691368	Rack
DL-SC-B-691375	Rack Supplement
DL-SC-B-691391	Power Junction Box
DL-SC-B-691538	S-250 Shelter Tie-Down
DL-SC-B-691547	Antenna Mounting Bracket
SC-C-691413	Ground Strap

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply:

Code of Federal Regulations, Title 29, Chapter XVII, Part 1910	Occupational Safety and Health Standards

(Applications for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, D.C. 20403.)

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3. REQUIREMENTS

3.1 First article. When specified in the contract or purchase order, the contractor shall furnish first article units in accordance with 4.3.

3.2 Construction (see 4.4). The kit assembly shall consist of the following units (including hardware), and shall be constructed in accordance with the kit assembly drawings.

a.	DL-SC-B-691368	Rack
	DL-SC-B-691375	Rack Supplement
	SC-C-691413	Ground Strap
b.	DL-SC-B-691547	Antenna Mounting Bracket
c.	DL-SC-B-691538	S-250 Shelter Tie-Down
d.	DL-SC-B-691391	Power Junction Box
	SC-C-691413	Ground Strap
e.	DL-SC-B-48521	Antenna Bracket

3.2.1 Finish. The kit assembly shall be finished in accordance with MIL-F-14072 and the kit assembly drawings. (see 4.4)

3.3 Environmental conditions. Specified components of the kit assembly shall meet the following service conditions:

3.3.1 Humidity. The power junction box shall be operable without degradation in specified electrical performance (see 3.4) and shall sustain no physical damage, during and after testing as specified in 4.6.1.

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3.3.2 Vibration. The kit assembly (power junction box, rack and rack supplement (assembled with dummy loads as shown in Figure 1)) shall withstand vibration, when tested in accordance with 4.6.2. Upon completion of the test, there shall be no evidence of mechanical and/or electrical degradation.

3.3.3 Salt fog. The antenna mounting bracket, S-250 shelter tie-down, power junction box and antenna bracket shall show no evidence of corrosion or clogging/binding of mechanical parts when tested as specified in 4.6.3.

3.3.4 Fungus. The power junction box, in the assembled and as ready-for-delivery condition shall provide no nutrients in material, coating, or contaminant form or support fungal growth when tested in accordance with 4.6.4. Only inherently fungus resistant grades of material in accordance with requirement 4 of MIL-STD-454 shall be used. Prior to delivery, all fungus nutrient contaminents or residues shall be removed.

3.3.5 Temperature (see 4.6.5).

3.3.5.1 High temperature.

(1) Operating. The power junction box shall be operable without degradation in electrical performance (see 3.4) at an ambient air temperature of 71°C (160°F).

(2) Storage and transportation. The power junction box shall withstand exposure to an ambient air temperature of 71°C (160°F).

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3.3.5.2 Low temperature.

(1) Operating. The power junction box shall be operable without degradation in electrical performance (see 3.4) at an ambient temperature of -45.6°C (-50°F).

(2) Storage and transporation. The power junction box shall withstand exposure to an ambient air temperature of -57°C (-70°F). When the ambient temperature is returned to -45.6°C (-50°F) there shall be no degradation in electrical performance (see 3.4).

3.3.6 Leakage (immersion). The power junction box shall show no evidence of leakage when tested in accordance with 4.6.6.

3.4 Electrical. The electrical requirements for the power junction box of the kit assembly shall be as follows:

3.4.1 Polarity. The input and output receptacles of the power junction box shall be wired in the following manner: (see 4.7.1)

Input receptacle:	Positive (+)	to pin A
	Negative (-)	to pin N
Output receptacle:	Positive (+)	to socket A
	Negative (-)	to socket B

3.4.2 Power-on. With 28 Vdc power supplied at the input receptacle, the indicator lamp of the power junction box shall light when the circuit breaker/switch is placed in the "ON" position, and shall be extinguished when the circuit breaker/switch is in the "OFF" position. (see 4.7.2)

3.4.3 Output voltage. The power junction box shall be wired such that when 28 Vdc power is supplied at the input receptacle (positive (+) to pin A and negative (-) to pin N), and the circuit breaker/switch is placed in the "ON" position, the voltage at each output receptacle (positive (+) to socket A and negative (-) to socket B) shall be 28 $\pm$.10 Vdc. (see 4.7.3)

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3.4.4 Overcurrent protection. The power junction box circuit breaker/switch shall be capable of interrupting all power to the junction box when the current exceeds 100 amps. (see 4.7.4)

3.5 Pull. The S-250 shelter tie-down shall not be permanently deformed when subjected to a load of 2000 pounds at each end and tested in accordance with 4.8.

3.6 Road test. The rack and rack supplement, antenna mounting bracket, power junction box and antenna bracket and the S-250 shelter tie-down shall be subjected to the road tests of 4.9.1 and 4.9.2, respectively, while installed on the GFP 1-1/4 ton commercial truck. The units, after the tests, shall show no evidence of mechanical deformation or degradation of electrical performance.

3.7 Interchangeability. Like units, assemblies, subassemblies, and replaceable parts shall be physically and functionally interchangeable without modification in any way. Individual items shall not be hand-picked for fit or performance. Reliance shall not be placed on any unspecified dimension, rating, characteristic, and no special tools shall be allowed. (See 4.10)

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3.8 Welding. Welded joints shall be as defined by the drawings of 3.2 and shall be such that grinding on the finish weld will be unnecessary except where grinding is specified on the drawings. Inert-gas-shielded arc welding (MIG) using helium, argon, or a mixture of the two shall be used for arc welding of aluminum. Welds shall conform to MIL-W-6858, Class B, and MIL-W-45205, Class B, as applicable; surfaces to be welded shall be cleaned in accordance with good commerical practice and shall be free of scale, paint, grease, and other foreign materials. Welds shall have thorough penetration and good fusion and shall be free of scabs, blisters, abnormal pock marks, cracks, voids, slag inclusions, and other harmful defects. Inert-gas-shielded arc welding shall be used for welding of aluminum except where spot welds are used. Welded assemblies shall be cleaned to remove any scale, oxidation products, and excess flux. Any acid used in cleaning shall be completely neutralized and removed. (see 4.4)

3.9 Nameplates and product markings. Nameplates and product markings shall conform to the drawings in 3.2 and MIL-M-13231. (see 4.4)

3.10 Workmanship. The kit assembly shall be manufactured and assembled in a thoroughly workmanlike manner in accordance with Requirement 9 of MIL-STD-454 and the applicable portions of 3.2 and 4.11.

3.11 Safety engineering. The kit assembly design criteria shall be in accordance with Title 29, Code of Federal Regulations, Chapter XVII, Part 1910, "Occupational Safety and Health Standards", MIL-STD-454, Requirement 1; and MIL-STD-1472, "Labeling" (para. 5.5) and "Hazards and Safety" (para. 5.13).

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsiblity for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may utilize his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection, does not include preparation for delivery. (see 4.3)

b. Inspections covered by subsidiary documents. (see 4.4)

c. Quality conformance inspections.

(1) Quality conformance inspection of equipment before preparation for delivery. (see 4.5)

(2) Quality conformance inspection of preparation for delivery. (see 4.5.5)

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4.3 First article. Unless otherwise specified in the contract or purchase order, the first article inspection shall be performed by the contractor. (See 3.1)

4.3.1 First article units. The contractor shall furnish two (2) units each of the rack and rack supplement, antenna mounting bracket, S-250 shelter tie-down, and antenna bracket and four (4) units of the power junction box.

4.3.2 First article inspection. The first article inspection shall consist of inspections specified in Table I and shall be performed in the order shown.

4.3.3 First article data. The first article test plan and test report(s) shall be as required in the contract or purchase order.

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TABLE I - FIRST ARTICLE INSPECTION

INSPECTIONS	REQT. PARA.	INSP. PARA.	SAMPLE SIZE: Rack and Rack Supplement	Antenna Mounting Bracket	S-250 Shelter Tie-Down	Power Junction Box 1/	Antenna Bracket
Inspection covered by subsidiary documents	3.2, 3.2.1 3.8, 3.9	4.4	Inspections shall be performed on all units, as applicable.				
Interchangeability	3.7	4.10	Inspections shall be performed on all units.				
Visual and mechanical	3.10	4.11	Inspections shall be performed on all units.				
Environmental							
Humidity	3.3.1	4.6.1	---	---	---	2	---
Vibration	3.3.2	4.6.2	2	---	---	2	---
Salt fog	3.3.3	4.6.3	---	2	2	2	2
Fungus	3.3.4	4.6.4	---	---	---	2	---
Temperature	3.3.5	4.6.5	---	---	---	2	---
Leakage (immersion)	3.3.6	4.6.6	---	---	---	2	---
Electrical							
Polarity	3.4.1	4.7.1	---	---	---	2	---
Power-on	3.4.2	4.7.2	---	---	---	2	---
Output voltage	3.4.3	4.7.3	---	---	---	2	---
Overcurrent protection	3.4.4	4.7.4	---	---	---	2	---
Pull	3.5	4.8	---	---	2	---	---
Road test 2/	3.6	4.9.1, 4.9.2	Inspections shall be performed on all units.				

1/ The four units of the power junction box shall be tested as follows:
   a. Two units shall be subjected to all first article inspections, as applicable, with the exception of the fungus test.
   b. Two units shall be subjected to the inspections of 4.4, 4.10, and 4.11 and shall then be subjected to the fungus test only. The fungus test shall commence at the start of first article inspection and shall run concurrently with all other tests.

2/ Installation of the units shall be accomplished in accordance with installation instructions supplied by the Government.

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4.4 Inspections covered by subsidiary documents. The following shall be inspected under the applicable subsidiary documents as part of the inspection required by this specification, and the inspection requirement specified in the contract or purchase order.

Item	Where Required
Construction	3.2
Finish	3.2.1
Welding	3.8
Nameplates and product markings	3.9
Safety engineering	3.11

4.5 Quality conformance inspection of equipment before preparation for delivery. The contractor shall perform the inspections specified in 4.4 and 4.5.1 through 4.5.4. This does not relieve the contractor of his responsibility for performing any additional inspection which is necessary to control the quality of the product and to assure compliance with all specification requirements.

4.5.1 Group A inspection. Each of the units of the kit assembly (see 3.2) shall be inspected for conformance to the inspections specified in Table II. Discrete lots shall be formed only from units that pass this inspection. Factors of lot composition not defined herein, or in the contract or purchase order, shall be in accordance with MIL-STD-105. Each discrete lot shall be subjected to sampling inspection, utilizing the procedures of MIL-STD-105, using the general inspection levels, and AQL's indicated in Table II.

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4.5.1.1 Order of inspection within Group A. Group A inspection shall be performed in the order shown in Table II.

TABLE II - GROUP A INSPECTION

INSPECTION	REQT. PARA.	INSP. PARA.	AQL (PERCENT DEFECTIVE) MAJOR	MINOR
Visual and mechanical (all units)	3.10	4.11	1.0	4.0
Electrical (power junction box only)				
Polarity	3.4.1	4.7.1)	1	*
Power on	3.4.2	4.7.2)	For	*
Output voltage	3.4.3	4.7.3)	The	*
Overcurrent protection	3.4.4	4.7.4)	Group	*

*All electrical defects are major

4.5.2 Group B inspection. Group B inspection shall normally be performed on inspection lots that have passed Group A inspection and on samples selected from units that have been subjected to and met the Group A inspection. This inspection shall conform to Table III and to the special inspection levels of Table I of MIL-STD-105.

4.5.2.1 Group B sampling plans. The Group B sampling plans for the AQL's listed in Table III shall be as follows:

AQL (percent defective)	Inspection Level
4.0	S-4
6.5	S-4

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4.5.2.2 Order of inspection within Group B. Group B inspection shall be performed in the order shown in Table III.

TABLE III - GROUP B INSPECTION

INSPECTION	REQT. PARA.	INSP. PARA.	AQL (percent defective)
Interchangeability (all units)	3.7	4.10	6.5
Leakage (immersion) (power junction box only)	3.3.6	4.6.6	4.0

4.5.3 Group C inspection. Group C inspection shall be performed on units that have passed Group A and Group B inspection. The inspection shall consist of the inspections specified in Table IV. Samples shall be selected in accordance with 4.5.3.1.

4.5.3.1 Sampling for Group C inspection. Two (2) units of the rack and rack supplement, five (5) units of the antenna mounting bracket, three (3) units of the S-250 shelter tie-down, four (4) units of the power junction box and two (2) units of the antenna bracket shall be selected at random for the inspections in Table IV.

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4.5.3.1.1 Frequency of sample selection. The frequency of sample selection shall be as shown in Table IV. The first sample shall be selected from the first quality conformance production lot.

4.5.3.2 Order of inspection within Group C. Group C inspection shall be performed in accordance with Note 1 of Table IV.

4.5.3.3 Group C failures. Actions required relative to Group C failures shall be as specified in the contract or purchase order.

4.5.4 Reinspection of conforming Group B and Group C sample units. Unless otherwise specified, sample units which have been subjected to and passed both Group B and Group C inspection may be accepted on the contract provided all damage is repaired and the sample units are resubjected to and pass Group A inspection.

4.5.5 Quality conformance inspection of preparation for delivery. Preparation for delivery shall be inspected in accordance with MIL-P-116 to determine conformance to the requirements of Section 5.

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TABLE IV - GROUP C INSPECTION

INSPECTION	REQT. PARA.	INSP. PARA.	SAMPLE SIZE: RACK AND RACK SUPPLEMENT	ANTENNA MOUNTING BRACKET	S-250 SHELTER TIE-DOWN	POWER JUNCTION BOX	ANTENNA BRACKET
ELECTRICAL TESTS, IN ACCORDANCE WITH 4.7, SHALL BE REPEATED FOLLOWING EACH TEST OF THE POWER JUNCTION BOX							
Temperature (High)	3.3.5.1	4.6.5.1	---	---	---	1 1/	---
Temperature (Low)	3.3.5.2	4.6.5.2	---	---	---		---
Salt Fog	3.3.3	4.6.3	---	1	1		1
Leakage (Immersion) 4/	3.3.6	4.6.6	---	---	---		---
Vibration	3.3.2	4.6.2	1	---	---		---
Fungus	3.3.4	4.6.4	---	---	---	1	---
Humidity	3.3.1	4.6.1	---	---	---	1	---
Road Test	3.6	4.9	1	4	1	1	1
Pull	3.5	4.8	---	---	1	---	---

SEE NOTES ON NEXT PAGE

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TABLE IV - GROUP C INSPECTION (CONT'D)

INSPECTION	REQT. PARA.	INSP. PARA.	SAMPLE FREQUENCY				
			RACK AND RACK SUPPLEMENT	ANTENNA MOUNTING BRACKET	S-250 SHELTER TIE-DOWN	POWER JUNCTION BOX	ANTENNA BRACKET
ELECTRICAL TESTS, IN ACCORDANCE WITH 4.7, SHALL BE REPEATED FOLLOWING EACH TEST OF THE POWER JUNCTION BOX							
Temperature (High)	3.3.5.1	4.6.5.1	---	---	---		---
Temperature (Low)	3.3.5.2	4.6.5.2	---	---	---		---
Salt Fog	3.3.3	4.6.3	---	2/	3/		3/
Leakage (Immersion) 4/	3.3.6	4.6.6	---	---	---	3/	---
Vibration	3.3.2	4.6.2	5/	---	---		---
Fungus	3.3.4	4.6.4	---	---	---		---
Humidity	3.3.1	4.6.1	---	---	---		---
Road Test	3.6	4.9	6/	2/	6/	6/	6/
Pull	3.5	4.8	---	---	7/	---	---

1/ The order of inspection for this sample shall be serially as listed, except as modified by Note 4. All other inspections shown in Table IV may be performed in parallel.
2/ Once every 10,000 units or fraction thereof.
3/ Once every 4,000 units or fraction thereof.
4/ The immersion test shall be performed prior to and after the vibration test of the power junction box.
5/ Once every 1,000 units or fraction thereof.
6/ Once every 3,000 units or fraction thereof.
7/ Once every 2,500 units or fraction thereof.

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4.6. Environmental conditions. The kit assembly shall be tested in accordance with the following environmental conditions:

4.6.1 Humidity. The power junction box shall be tested in accordance with Method 507, Procedure III, of MIL-STD-810. In step 6, measurements shall be taken during the last 5 hours of the last cycle. Inability of the power junction box the electrical requirements of 3.4 and any evidence of physical damage constitutes failure of this test.

4.6.2 Vibration.

4.6.2.1 Pre-test conditions. Inspect all test items mechanically. The rack and rack supplement shall be tested with dummy loads simulating actual equipment installation (see Figure 1). Operate the power junction box and record preperformance electrical test measurements. Test item operation shall be evidenced by the ability of the power junction box to meet the electrical requirements of 3.4.

4.6.2.2 Mounting techniques. The kit assembly (power junction box, and rack and rack supplement only) shall be attached to the vibrator exciter table by its normal mounting means or by means of a rigid fixture capable of transmitting the vibration conditions specified herein. Precautions shall be taken in the establishment of mechanical interfaces to minimize the introduction of undesirable responses in the test setup. Whenever possible, the test load shall be distributed uniformly on the vibrator exciter table in order to reduce, to the smallest amount, effects of unbalanced loads. Vibration amplitudes and frequencies shall be measured by techniques that will not significantly affect test-item input control or response. The input control sensing device(s) shall be rigidly attached to the vibration table or to the intermediate structure, if used, at, or as near as possible to, the attachment points of the test items.

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4.6.2.3 Cycling. The power junction box, and rack and rack supplement shall be vibrated along each of their three (3) mutually perpendicular axis in accordance with Test Level, Frequency Range, and Time Schedule specified. The frequency of the applied vibration shall be swept logarithmically over the specified frequency range.

Test Level:	Vibration frequency of 5.5 Hz to 200 Hz at an acceleration level of 1.5 Gs
Frequency Range:	5.5-200-5.5 Hz
Time Schedule:	3 hours per axis
Swept Rate:	5.5-200-5.5 Hz in twelve (12) minutes

The power junction box shall be energized and operated during not less than 20 minutes of each hour and voltage readings shall be recorded for both input and output receptacles (one set of readings for each 5 minutes operated). At the conclusion of the test, visually inspect the test items for any evidence of mechanical damage and operate the power junction box and compare the results with the preperformance and operational electrical test data. (See 3.3.2)

4.6.3 Salt fog. The antenna mounting bracket, S-250 shelter tie-down, power junction box and antenna bracket shall be subjected to the test of Method 509, Procedure I of MIL-STD-810. Operation of the units shall not be required (except for moving mechanical parts) and failure criteria shall be limited to corrosion of finishes and metals and any clogging or binding of mechanical parts. Corrosion shall be defined as any visible degradation of the equipment surfaces that can be attributed to flakey, pitted, blistered, or otherwise loosened finish or metal surfaces; or exfoliation (6.5.2) of metal. (See 3.3.3)

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4.6.4 Fungus. The power junction box, in the assembled and as-ready-for-delivery condition shall be subjected to the test of Method 508 of MIL-STD-810. The power junction box shall not be specially cleaned for this test, except for the cleaning it receives during or after production. If, after assembly, it becomes necessary for the manufacturer to remove accumulated production and handling contaminants by cleaning prior to packaging and delivery, the power junction box shall not be tested for three days after cleaning to permit complete evaporation of the cleaning compound. All enclosed or gasketed assemblies shall be opened and the interior sprayed with the specified mixed spore suspension. After 14 and 28 days of test, the control items in accordance with 3.1.3 and 3.1.4 of Method 508 of MIL-STD-810 shall show profuse growth over at least 50 percent of the area of the control items. After 28 days, based upon visual examination, the power junction box shall pass if it shows no more than only sparse microbial growth (6.5.3) as evidenced by growth colonization (6.5.6) (which includes branching (6.5.1) and sporulation (6.5.4) on or within each cubic foot or fraction thereof of equipment assembly volume) with restricted tubular growth development (6.5.5) in an area of 1 to 10 percent or less of the total area and no more than six unrelated minute colonies with mycelial development in areas only in other than critical circuit portions; such as terminal spacing, printed circuit boards, etc., with sparse growth due to random contamination or traces of unmixed material ingredients. No operational test is required after the fungus test. The power junction box shall fail if it exhibits more than the growth specified. The test shall be repeated if the control items fail to show profuse growth after 14 and 28 days of test. (see 3.3.4)

4.6.5 Temperature (see 3.3.5).

4.6.5.1 High temperature. The power junction box shall be tested in accordance with Method 501, Procedure II, of MIL-STD-810. For step 4, the chamber temperature shall be 71°C (160°F). For step 7, the chamber temperature shall be adjusted to 71°C (160°F) for electrical operation of the power junction box.

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4.6.5.2 Low temperature. The power junction box shall be tested in accordance with Method 502, Procedure I, of MIL-STD-810. For step 2, the storage temperature shall be -57°C (-70°F) and shall be maintained for a period of not less than 2 hours following stabilization of the test item. For step 4, the chamber temperature shall be adjusted to -45.6°C (-50°F) for electrical operation of the power junction box.

4.6.6 Leakage (immersion). The power junction box shall be tested in accordance with Method 512, Procedure I, of MIL-STD-810. There shall be no evidence of leakage after the test. (see 3.3.6)

4.7 Electrical. All electrical tests may be conducted simultaneously, provided all requirements of 4.7.1 through 4.7.4 are met.

4.7.1 Polarity. Polarity for the power junction box shall be tested in the following manner (see 3.4.1):

1. Probes shall be placed between pin A of the input receptacle and socket A of the output receptacles (individually). Electrical continuity shall be indicated.

2. Probes shall be placed between pin N of the input receptacle and socket B of the output receptacles (individually). Electrical continuity shall be indicated.

4.7.2 Power-on. The power junction box shall be connected to a power source capable of supplying 28 Vdc. After connected to the source, the circuit breaker/switch of the power junction box shall be placed in the "ON" position and the power-on indicator lamp shall light. The circuit breaker/switch shall then be placed in the "OFF" position and the indicator lamp shall be extinguished. This ON/OFF cycle shall be repeated 5 times with the same results. (see 3.4.2)

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4.7.3 Output voltage. The power junction box shall be connected to a power source capable of supplying 28 Vdc. After connected to the source, the voltage on the input and output receptacles shall be measured and recorded. The voltage at each receptacle shall be 28 ±0.10 Vdc. (See 3.4.3)

4.7.4 Overcurrent protection. The power junction box shall be connected to a power source capable of supplying 28 Vdc and shall be loaded in such a manner as to draw 110, 115 and 120 amps (a minimum of six output receptacles shall be used to apply the load). The circuit breaker/switch shall interrupt all power to the junction box when these currents are reached (time duration of overload required for breaker activation shall be obtained from the breaker manufacturer's curves). (See 3.4.4)

4.8 Pull. With the S-250 shelter tie-down installed in its normal position using required mounting hardware (see 3.2), a load of 2,000+50-0 pounds shall be applied perpendicular to each end at the eye bolts. The direction of the load shall be vertical and the load shall be applied for a period of one hour. At the end of the test, there shall be no evidence of permanent deformation. (See 3.5)

4.9 Road test (see 3.6).

4.9.1 Rack and rack supplement, antenna mounting bracket, power junction box, and antenna bracket. The rack and rack supplement (loaded in accordance with Figure 1), the antenna mounting bracket, the power junction box, and the antenna bracket shall be installed on the GFP 1-1/4 ton commercial truck. The truck shall then be road-tested on roadways approved by the Government. The road test consists of the following:

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4.9.1 (Cont'd)

1. 30 miles, at 55 miles per hour (mph) on improved roadways.

2. 20 miles, at 55 mph on unimproved road surfaces.

3. Upon completion of the test, the rack and rack supplement, the mounting, the power junction box, and the antenna bracket shall be examined for evidence of physical deformation and electrical performance.

4.9.2 S-250 shelter tie-down. A GFP S-250( )/G shelter shall be installed on the GFP 1-1/4 ton commercial truck using normal shelter slings and the S-250 shelter tie-down. The truck shall then be road-tested on roadways approved by the Government. The road test consists of the following:

1. 30 miles, at 55 mph on improved roadways.

2. 20 miles, at 55 mph on unimproved road surfaces.

3. Upon completion of the test, the S-250 shelter tie-down shall be examined for evidence of physical deformation.

4.10 Inspection for dimensional interchangeability (see 3.7). The dimensions listed below shall be gaged or measured to determine compliance with the physical interchangeability requirements of 3.7. When the listed dimensions are not within specified or design limits, it shall be considered a major defect.

a. External and internal dimensions of covers, and insertable assemblies and parts, when such dimensions affect mating of parts.

b. Size and mounting dimensions of brackets and mounting plates.

c. Location of hinges and fasteners on separable parts and assemblies which must mate, such as covers, parts and mountings.

4.11 Visual and mechanical inspection. The kit assembly units listed in 3.2 shall be examined for the defects listed in MIL-STD-252 and for compliance with Requirement 1 of MIL-STD-454. (see 3.10)

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5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be level A or C, as specified. (see 6.2)

5.1.1 Level A.

5.1.1.1 Cleaning. Kit assembly shall be cleaned in accordance with process C-1 of MIL-P-116.

5.1.1.2 Drying. Kit assembly shall be dried in accordance with the applicable procedure of MIL-P-116.

5.1.1.3 Preservation application. None required.

5.1.1.4 Unit packaging. Unit packaging shall be in accordance with the methods prescribed in MIL-P-116 as specified herein.

5.1.1.4.1 Technical literature. Each technical literature shall be packaged method IC-1.

5.1.1.4.2 Kit assembly. Each kit assembly component shall be individually packaged method III as follows:

5.1.1.4.2.1 Power junction box. Cushion each unit on all surfaces with cells or pads or both fabricated of fiberboard conforming to PPP-F-320, type CF, class weather-resistant, variety SW, grade W5c, designed to protect all projections and absorb the shock of impact in handling and transit. As an alternate, cushioning material may be fabricated of plastic molding material conforming to MIL-P-19644 or polyurethane foam conforming to MIL-P-26514. Place the cushioned unit within a close-fitting fiberboard box conforming to PPP-B-636, W5c. Place the technical literature, packaged as specified in 5.1.1.4.1, on top of the contents, directly under the lid of the box. Close the box in accordance with the appendix of the box specification.

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5.1.1.4.2.2 Antenna mounting bracket. Cushion each unit with cellulosic material conforming to PPP-C-843, type II. Place the cushioned unit within a close-fitting fiberboard box conforming to PPP-B-636, W5c. Close the box in accordance with the appendix of the box specification.

5.1.1.4.2.3 S-250 shelter tie-down. Nest S-250 shelter tie-down and cushion with paperboard conforming to PPP-P-291. Contain cushioning material with tape conforming to PPP-T-97, type II, class A.

5.1.1.4.2.4 Rack. Each rack section shall be packaged disassembled as follows:

5.1.1.4.2.4.1 Base plate. Each base plate shall be cushioned on all faces with paperboard conforming to PPP-P-291. Secure paperboard by stapling.

5.1.1.4.2.4.2 Rack. Each component of the rack assembly as specified in Drawing No. DL-SC-B-691368 shall be cushioned on all faces with paperboard conforming to PPP-P-291. Secure paperboard by stapling. Unitize these components by banding together with strapping conforming to QQ-S-781, type I, finish A.

5.1.1.4.2.4.3 Rack supplement. Each component of the rack supplement as specified in Drawing No. DL-SC-B-691375 shall be cushioned on all faces with paperboard conforming to PPP-P-291. Secure paperboard by stapling. Unitize these components by banding together, with strapping conforming to QQ-S-781, type 1, finish A.

5.1.1.4.2.5 Antenna bracket. Cushion each unit with cellulosic material conforming to PPP-C-843, Type II. Place the cushioned unit within a close-fitting fiberboard box conforming to PPP-B-636, W5c. Close the box in accordance with the appendix of the box specification.

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5.1.1.5 Hardware. Nuts, bolts, screws, etc., necessary for installation/assembly shall be inserted in cotton bags of suitable size conforming to PPP-B-20. Each bag shall be securely tied to each kit assembly unit.

5.1.2 Level C. Kit assembly shall be preserved and packaged in a manner that will afford adequate protection against physical and environmental damage during shipment, handling and limited intransit storage.

5.2 Packing. Packing shall be level A, B or C, as specified. Shipping containers for all levels shall be capable of stacking and supporting superimposed loads during shipment and storage without damaging the container(s) or its contents (see 6.2).

5.2.1 Level A. Each kit assembly packaged as specified in 5.1, shall be packed within a close-fitting box conforming to PPP-B-621, class 2, style 4. Skids, 3 x 4 inches, laid flat shall be applied in accordance with the container specification or appendix thereto except that metal strapping shall conform to QQ-S-781, type I, finish A.

5.2.2 Level B. Each kit assembly packaged as specified in 5.1 shall be packed as specified in 5.2.1, except that box shall be class 1.

5.2.3 Level C. Each kit assembly packaged as specified in 5.1, shall be packed in shipping containers that comply with Federal or Military specifications, in a manner that will afford adequate protection to the package and its contents against physical and environmental deterioration and damage, during limited shipment, handling and intransit storage.

5.3 Marking. In addition to any special marking required by the contract or order, interior packages and exterior shipping containers shall be marked in accordance with MIL-STD-129.

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6. NOTES

6.1 Intended use.

a. The rack and rack supplement, antenna mounting bracket, power junction box, and antenna bracket of the kit assembly provide for installation of the following standard Army radios, in various combinations, on the 1-1/4 ton commercial truck:

AN/VRC-12, 24, 43, 46, 47, 49, 53, 64
AN/GRC-106, 125, 160

b. The S-250 shelter tie-down will permit installation of the S-250 shelter on the 1-1/4 ton commercial truck.

6.1.1 Compatibility. The kit assembly must be compatible with the following items:

a. Trucks, commercial; 1-1/4 ton, XM861 Series, 60 ampere and 100 ampere, 28 volt Generating System Kit.

b. Trucks, commercial; 1-1/4 ton, XM861 Series, Cargo Box Cover Kit.

c. Trucks, commercial; 1-1/4 ton, XM861 Series, Troop Seat Kit.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification and any amendment thereto.

b. Level of preservation and packaging and packing (see Section 5).

c. When rough handling and functional tests are required.

d. Quantity of First Article units.

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6.2 (Cont'd)

e. Marking and shipping of samples.

f. Place of final inspection.

g. A list of Government property to be loaned for testing purposes.

h. First article and Group C inspection of the rack and rack supplement requires the use of contractor-fabricated dummy loads as specified in the bid request and contract.

6.3 Group C inspection. Approval to ship may be withheld, at the discretion of the Government, pending the decision from the contracting officer on the adequacy of corrective action. (see 4.5.3)

6.3.1 Location of inspection. It is desirable that the inspection be performed at a location that will minimize handling after this inspection is completed.

6.4 Verification inspection. Verification by the Government will be limited to the amount deemed necessary to determine compliance with the contract and will be limited in severity to the quality assurance provisions established in this specification and the contract.

6.5 Definitions.

6.5.1 Branching. Branching is a connected arrangement of filaments (hyphae) formed by shoots or secondary stems growing from the main stem or filament (hypha).

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6.5.2 Exfoliation. Exfoliation is corrosion along the grain boundaries of the metal resulting in the peeling or separating, or both, of successive layers of the metal. The appearance resembles loose book pages or onion skin peeling.

6.5.3 Microbial growth. Microbial growth is the growth of very minute organisms. Such organisms when present in large numbers may provide a colony visible to the naked eye.

6.5.4 Sporulation. Sporulation is the formation of minute unicellular reproductive or dormant bodies, called spores.

6.5.5 Tubular germination. Tubular germination is partial growth by the production of hyphae, which are tubular shaped fungal filaments. Tubular germination constitutes restricted individual spore growth not proceeding to colonization.

6.5.6 Growth colonization. Growth colonization is a mass of individual plants, generally of one species, living together; or a group of hyphae which is formed from one spore or cell and may be one individual plant. Colonization which completely covers the surface of the nutrient material constitutes abundant growth.

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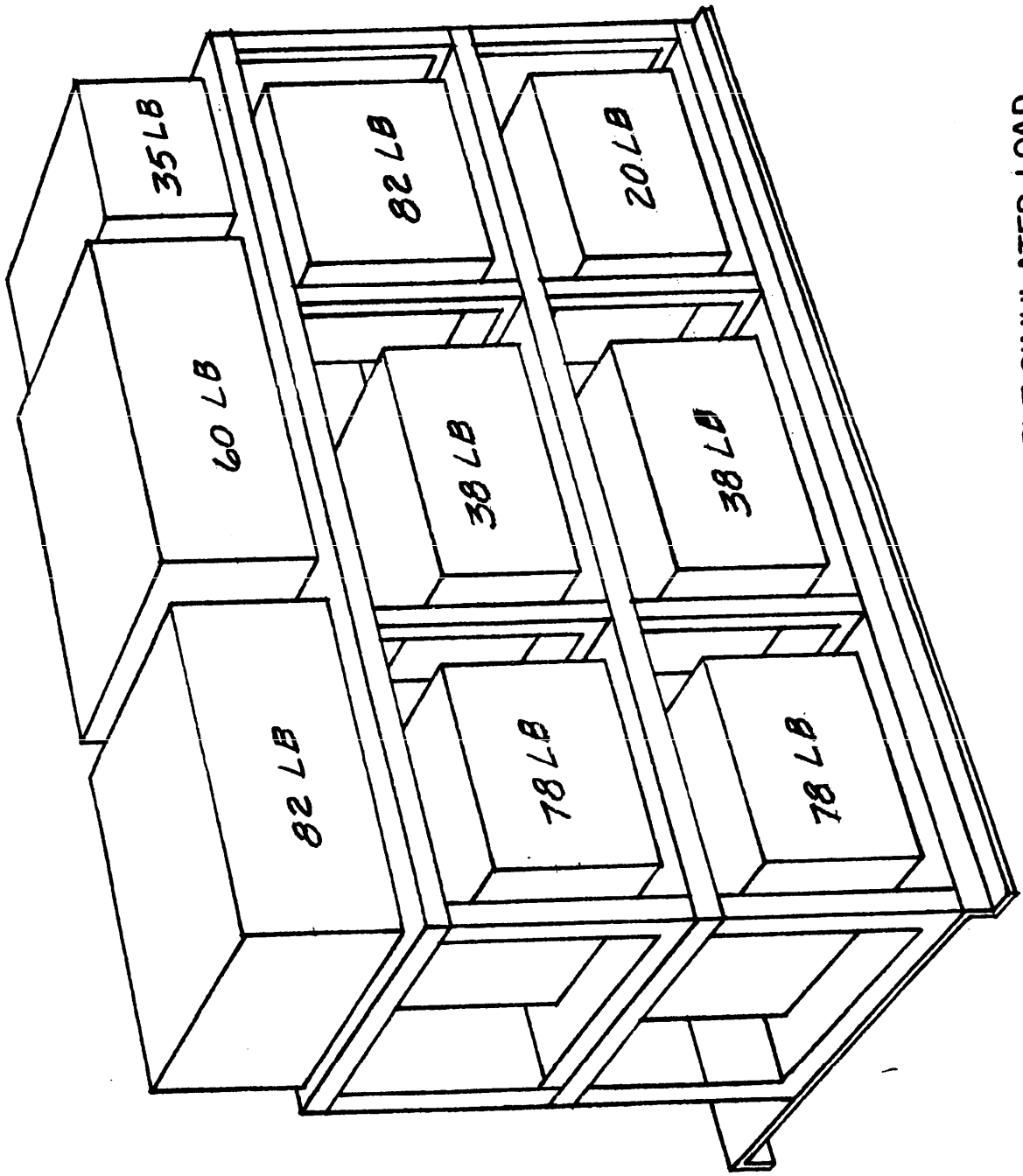
ARMY-EL

Preparing activity

ARMY-EL

Project Nr. 5820-A653

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RACK AND RACK SUPPLEMENT SIMULATED LOAD
FIGURE 1

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION MIL-K-49054(EL) KIT, INSTALLATION OF COMMUNICATIONS EQUIPMENT IN 1-1/4 TON COMMERCIAL TRUCK

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

ESC-FM 1068-68

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Headquarters
U.S.Army Electronics Command
Fort Monmouth, New Jersey 07703

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